Topic 6: SVM in the Dual Form

Projected Gradient (Week 6) versus Barrier Method (Week 5)

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Abstract

We solve the dual of a soft-margin kernel SVM on the two-moons dataset with two methods: our own projected gradient (PG) method from Week 6 (proximal gradient with the indicator of the feasible set as the nonsmooth term, projection computed by bisection), and the log-barrier interior-point method of Week 5 as the baseline. Both are validated against `sklearn.svm.SVC`. At $C = 1,\ \gamma = 1$ both reach the required accuracy and recover the same 63 support vectors. The barrier method needs 87 Newton steps (0.25 s) while PG needs 7302 iterations (4.3 s). We then study two personal factors: the penalty parameter $C \in \{0.1, 0.3, 1, 3\}$ and the kernel width $\gamma \in \{0.5, 1, 2, 5\}$.

1 Problem and data

Given points $x_i \in \mathbb{R}^2$ with labels $y_i \in \{-1, +1\}$, the dual of the soft-margin SVM with an RBF kernel is the quadratic program

$$\min_{\boldsymbol{\alpha} \in \mathbb{R}^n} \ \tfrac{1}{2}\,\boldsymbol{\alpha}^\top Q \boldsymbol{\alpha} - \mathbf{1}^\top \boldsymbol{\alpha} \quad \text{s.t.} \quad 0 \le \alpha_i \le C, \quad \boldsymbol{y}^\top \boldsymbol{\alpha} = 0, \qquad Q_{ij} = y_i y_j\, e^{-\gamma \|x_i - x_j\|^2}. \tag{1}$$

The matrix Q is positive semidefinite, so (1) is convex. It has a box constraint and one linear equality, which makes it a standard test case for both a projection method and an interior-point method.

Data. `make_moons(n_samples=200, noise=0.25, random_state=0)`, labels mapped to $\{-1, +1\}$ (100 points per class). Defaults: $C = 1$, $\gamma = 1$.

Reference. `SVC(kernel='rbf', tol=1e-12)` with $\alpha_i = |$`dual_coef_`$_i|$ at `support_` indices and 0 elsewhere. Its objective value $f^\star$ is used to define the dual gap $|f(\boldsymbol{\alpha}) - f^\star|$.

2 Methods

2.1 Projected gradient (own method, Week 6)

Lecture 6 treats composite problems $\min F = f + g$ with f convex and differentiable, and g convex, closed, with a cheap proximal operator. Problem (1) fits this form with

$$f(\boldsymbol{\alpha}) = \tfrac{1}{2}\boldsymbol{\alpha}^\top Q \boldsymbol{\alpha} - \mathbf{1}^\top \boldsymbol{\alpha}, \qquad g = \iota_{\mathcal{C}}, \qquad \mathcal{C} = \{\boldsymbol{\alpha} :\ 0 \le \alpha_i \le C,\ \boldsymbol{y}^\top \boldsymbol{\alpha} = 0\}.$$

Here $\nabla f(\boldsymbol{\alpha}) = Q\boldsymbol{\alpha} - \mathbf{1}$ is Lipschitz with $L = \lambda_{\max}(Q)$ (the Hessian is the constant matrix Q). For the indicator of a convex set the proximal operator is the projection, $\operatorname{prox}_{t\iota_{\mathcal{C}}} = P_{\mathcal{C}}$ for every $t > 0$, so the proximal gradient step with $t = 1/L$ is the projected gradient step

$$\boldsymbol{\alpha}^{k+1} = P_{\mathcal{C}}\big(\boldsymbol{\alpha}^k - t\,(Q\boldsymbol{\alpha}^k - \mathbf{1})\big), \qquad t = \frac{1}{\lambda_{\max}(Q)}, \qquad \boldsymbol{\alpha}^0 = 0.$$

Projection. The set $\mathcal{C}$ is the intersection of a box and a hyperplane; separately these have closed forms (clipping, an affine formula), but the intersection needs a one-dimensional search. The projection problem $\min_u \frac{1}{2}\|u - v\|^2$ s.t. $0 \le u \le C$, $\boldsymbol{y}^\top u = 0$ has stationarity conditions that, with a multiplier ν for the equality, give

$$P_{\mathcal{C}}(v) = \mathrm{clip}(v - \nu\boldsymbol{y},\, 0,\, C),$$

where ν is the root of $\varphi(\nu) = \boldsymbol{y}^\top \mathrm{clip}(v - \nu\boldsymbol{y}, 0, C)$. Each term $y_i \,\mathrm{clip}(v_i - \nu y_i, 0, C)$ is non-increasing in ν (because $y_i^2 = 1$), so φ is non-increasing and the root is found by bisection: 50 steps on $[-50, 50]$, final bracket width $\approx 10^{-13}$.

Stopping and guarantees. As in Lecture 6, the stopping quantity is the gradient mapping $\|\boldsymbol{\alpha}^k - \boldsymbol{\alpha}^{k+1}\|/t < 10^{-6}$ (with a limit of `max_iter`$= 20000$). It vanishes exactly at solutions, since $\boldsymbol{\alpha}^\star$ minimizes F if and only if $\boldsymbol{\alpha}^\star = P_{\mathcal{C}}(\boldsymbol{\alpha}^\star - t\nabla f(\boldsymbol{\alpha}^\star))$. For $t = 1/L$ the theory gives (i) sufficient decrease, $F(\boldsymbol{\alpha}^{k+1}) \le F(\boldsymbol{\alpha}^k) - \frac{L}{2}\|\boldsymbol{\alpha}^{k+1} - \boldsymbol{\alpha}^k\|^2$, so F never increases; and (ii) the global rate

$$F(\boldsymbol{\alpha}^k) - F^\star \;\le\; \frac{L\,\|\boldsymbol{\alpha}^0 - \boldsymbol{\alpha}^\star\|^2}{2k}, \tag{2}$$

which is sublinear, $O(1/k)$. We check both properties on our runs in Section 4.

2.2 Barrier method (baseline)

Lecture 5 replaces the complementarity condition $\mu_j s_j = 0$ by $\mu_j s_j = \tau$ and follows the central path as $\tau \downarrow 0$. The barrier method does this by eliminating μ. We write the box as $G\boldsymbol{\alpha} \le h$ with $G = [-I;\, I]$, $h = [0;\, C\mathbf{1}]$ (so $p = m = 2n = 400$ inequalities) and keep $\boldsymbol{y}^\top\boldsymbol{\alpha} = 0$ as an equality; the slack is $s = h - G\boldsymbol{\alpha}$ and the barrier weight is $\tau = 1/t$. For each t the centering problem

$$\min_{\boldsymbol{\alpha}}\; t\Big(\tfrac{1}{2}\boldsymbol{\alpha}^\top Q\boldsymbol{\alpha} - \mathbf{1}^\top\boldsymbol{\alpha}\Big) - \sum_{j=1}^{m} \log\big(h_j - G_j\boldsymbol{\alpha}\big) \quad \text{s.t.} \quad \boldsymbol{y}^\top\boldsymbol{\alpha} = 0$$

is solved by Newton's method with equality constraints. Each step solves the KKT system

$$\begin{pmatrix} H & \boldsymbol{y} \\ \boldsymbol{y}^\top & 0 \end{pmatrix} \begin{pmatrix} \Delta\boldsymbol{\alpha} \\ w \end{pmatrix} = \begin{pmatrix} -g \\ 0 \end{pmatrix}$$

of size $n + 1 = 201$ (H is the Hessian of the barrier function, g its gradient), followed by backtracking line search (Armijo constant 0.01, factor 0.5) that keeps every iterate strictly feasible. The start is $\boldsymbol{\alpha}^0 = \frac{C}{2}\mathbf{1}$, which is strictly feasible and satisfies $\boldsymbol{y}^\top\boldsymbol{\alpha}^0 = 0$ because both classes have 100 points. The barrier parameter starts at $t_0 = 1$ and is multiplied by 10 after each centering. We stop when $m/t \le 10^{-8}$, which, by the duality-gap theorem of Lecture 5 ($f(x_\tau) - f^\star \le p\tau$ for convex problems), certifies $f(\boldsymbol{\alpha}_t) - f^\star \le m/t$. This happens at $t = 10^{11}$, i.e. after 12 outer iterations, with certified gap 4×10^{-9}. Newton's method inside a centering step stops when $\lambda^2/2 \le 10^{-9}$ or after 100 steps.

2.3 Quality measures

- **Dual gap** $|f(\boldsymbol{\alpha}) - f^\star|$ against `SVC`. Targets: $\le 10^{-6}$ for PG, $\le 10^{-8}$ for barrier.
- **KKT violation:** the maximum over the following. For $\alpha_i = 0$: $\max(0, 1 - y_i f(x_i))$. For free $0 < \alpha_i < C$: $|y_i f(x_i) - 1|$. For $\alpha_i = C$: $\max(0, y_i f(x_i) - 1)$. Plus $|\boldsymbol{y}^\top\boldsymbol{\alpha}|$ and the box violation, with tolerance 10^{-5} for classifying α_i as 0 or C. Here $f(x) = \sum_i \alpha_i y_i k(x_i, x) + b$. The bias b is not part of the dual; it is recovered as the mean of $y_i - \sum_j \alpha_j y_j k(x_i, x_j)$ over the free support vectors.
- **Support vectors (SV):** $\#\{\alpha_i > 10^{-6}\}$.
- **Cost:** PG iterations; for the barrier method, total Newton steps. Wall-clock time is also reported.

3 Verification at $C = 1$, $\gamma = 1$

Table 1 compares the two methods with `SVC`. Both reach the required accuracy and find exactly the same 63 support vectors as `SVC`. All three solutions classify 94% of the training points correctly. The equality constraint is satisfied to machine precision. The barrier solution is slightly above the optimum by construction, since the iterates are interior and the gap is bounded by m/t.

Table 1: Main comparison at $C = 1$, $\gamma = 1$.

	Projected gradient	Barrier method	`SVC` (reference)
Dual objective	-46.70335586519	-46.70335586329	-46.70335586519
Dual gap $\lvert f - f^\star \rvert$	3.1×10^{-12}	1.9×10^{-9}	–
Target gap	$\le 10^{-6}$	$\le 10^{-8}$	–
KKT violation	6.0×10^{-7}	3.4×10^{-10}	1.6×10^{-7}
$\lvert \boldsymbol{y}^\top \boldsymbol{\alpha} \rvert$	2.0×10^{-13}	2.6×10^{-15}	8.9×10^{-16}
Iterations	7302 PG iterations	87 Newton (12 outer)	–
Time, s	4.29	0.25	–
Support vectors	63	63	63
Training accuracy	0.94	0.94	0.94

Figure 1 shows the dual gap against the number of Newton steps / PG iterations. The barrier method brings the gap to about 10^{-8} after roughly 80 Newton steps. PG needs several thousand iterations: it has a long slow phase (gap 10^{-3}-10^{-4}), then a quick drop once the active set of the box constraints settles, followed by fast linear convergence.

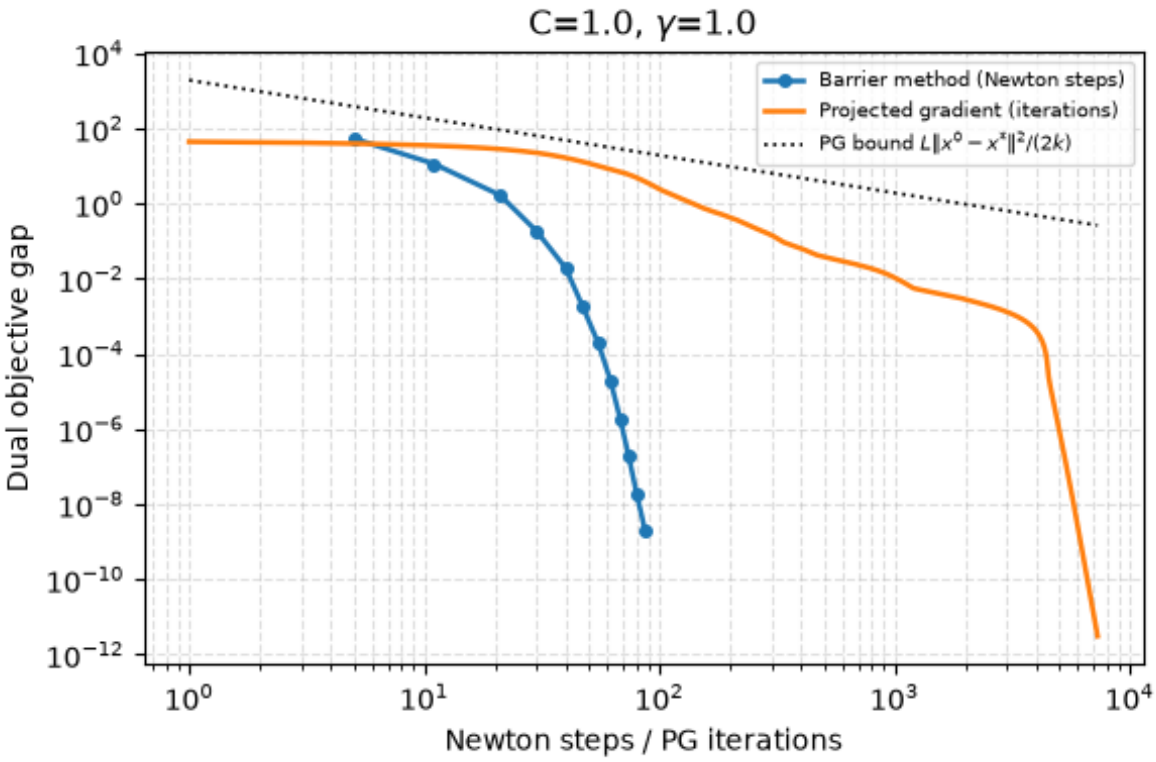


Figure 1: Dual gap versus iterations (log-log).

4 Check against the theory of Lecture 6

We tested the guarantees of Section 2.1 on all seven distinct runs (the $C = 1$, $\gamma = 1$ run is shared by both parameter sweeps).

- **Monotone decrease.** The objective value decreased (did not increase) at every one of the PG iterations in every run, as sufficient decrease predicts.
- **Rate bound** (2)**.** With $\boldsymbol{\alpha}^0 = 0$ and $\boldsymbol{\alpha}^\star$ taken from `SVC`, the bound $L\|\boldsymbol{\alpha}^\star\|^2/(2k)$ holds at every iteration k in every run. It is drawn as a dotted line in the convergence figures.
- **Fixed point.** At the `SVC` solution the gradient mapping is between 4×10^{-8} and 7×10^{-7} in all runs, i.e. below the stopping tolerance. So the reference solution satisfies the optimality condition $\boldsymbol{\alpha}^\star = P_{\mathcal{C}}(\boldsymbol{\alpha}^\star - t\nabla f(\boldsymbol{\alpha}^\star))$ up to the solver tolerance of `SVC`.

The bound is very pessimistic as an iteration estimate. Solving $L\|\boldsymbol{\alpha}^\star\|^2/(2k) \le 10^{-6}$ for k gives 10^7-10^{10} iterations, while PG stopped after 10^4-3×10^4 (Table 2). Lecture 6 states that practice is often much faster than the worst-case bound, and the plateau followed by a sharp drop visible in the figures resembles the $\gamma = 3$ example of the lecture, where the support is identified first and convergence is fast afterwards. We did not verify this explanation by tracking the active set.

Table 2: Constants of the bound (2) and actual PG cost. Actual iterations are those at which the residual fell below 10^{-6}, taken from runs with a larger iteration limit where the 20000 limit was reached ($C = 0.1$, $C = 3$, $\gamma = 2$).

Run	$L = \lambda_{\max}(Q)$	$\|\boldsymbol{\alpha}^\star\|^2$	k required by bound for 10^{-6}	Actual PG iterations
$C = 0.1, \gamma = 1$	68.9	1.18	4.1×10^7	21775
$C = 0.3, \gamma = 1$	68.9	7.13	2.5×10^8	10806
$C = 1, \gamma = 1$	68.9	55.8	1.9×10^9	7302
$C = 3, \gamma = 1$	68.9	387	1.3×10^{10}	31015
$C = 1, \gamma = 0.5$	99.8	68.6	3.4×10^9	14001
$C = 1, \gamma = 2$	43.5	46.5	1.0×10^9	22531
$C = 1, \gamma = 5$	21.5	43.6	4.7×10^8	7923

5 Personal factor: penalty parameter C ($\gamma = 1$)

Table 3: Effect of C. * PG stopped at the 20000-iteration limit.

C	0.1	0.3	1	3
Support vectors (all three methods)	122	86	63	49
Free SV ($0 < \alpha < C$)	8	9	10	9
Margin $1/\sqrt{\boldsymbol{\alpha}^\top Q \boldsymbol{\alpha}}$	0.407	0.309	0.216	0.151
Training accuracy	0.910	0.940	0.940	0.945
PG iterations	20000*	10806	7302	20000*
PG dual gap	7.2×10^{-11}	3.1×10^{-12}	3.1×10^{-12}	2.1×10^{-4}
PG KKT violation	1.05×10^{-6}	6.3×10^{-7}	6.0×10^{-7}	1.6×10^{-3}
PG time, s	11.40	7.24	4.29	11.50
Barrier Newton steps (outer)	178 (12)	182 (12)	87 (12)	93 (12)
Barrier dual gap	1.9×10^{-9}	1.9×10^{-9}	1.9×10^{-9}	1.9×10^{-9}
Barrier KKT violation	1.8×10^{-9}	3.3×10^{-10}	3.4×10^{-10}	5.8×10^{-11}
Barrier time, s	0.57	0.43	0.25	0.23

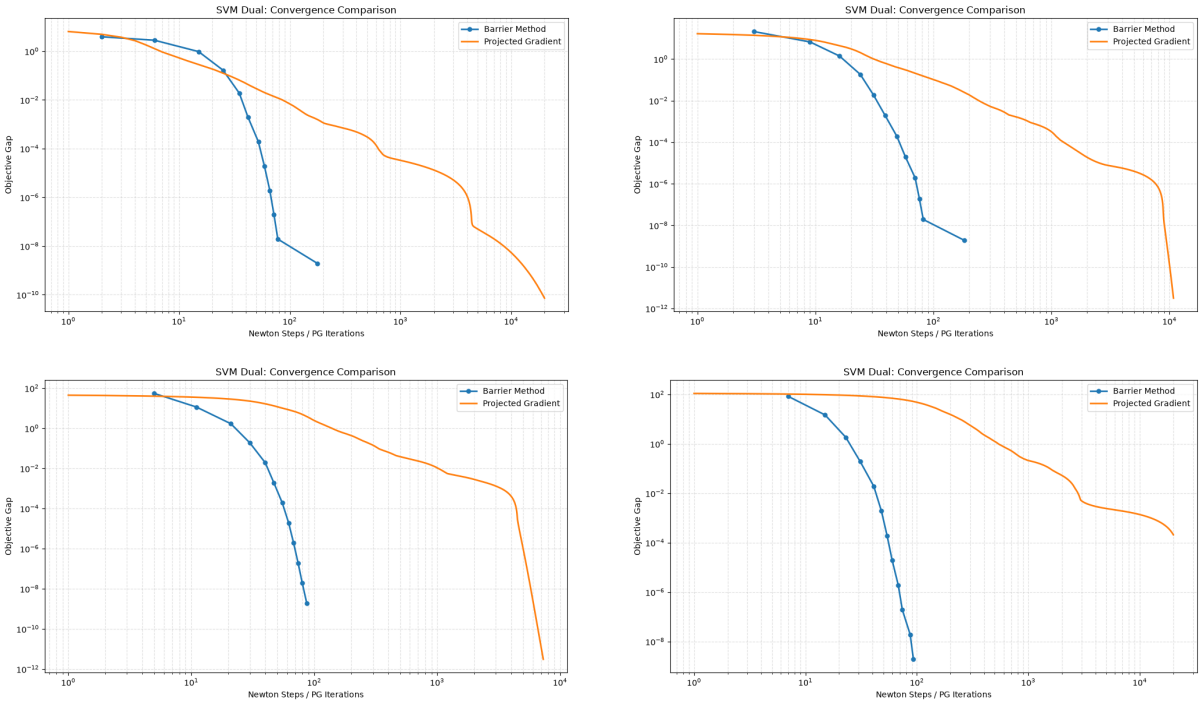


Figure 2: Convergence for $C \in \{0.1, 0.3, 1, 3\}$ ($\gamma = 1$).

Observations.

- *Support vectors and margin.* Increasing C makes margin violations more expensive. The number of support vectors drops from 122 to 49, and the margin $1/\|w\|$ shrinks from 0.41 to 0.15. At small C most support vectors sit at the upper bound $\alpha_i = C$ (only 8 of 122 are free). Training accuracy rises slightly, from 0.91 to 0.945. This is training accuracy only; we did not measure generalization.
- *PG cost.* For $C = 1$ PG needs 7302 iterations. At $C = 3$ it had not converged after

20000 iterations (gap 2.1×10^{-4}, KKT violation 1.6×10^{-3}); the topic sheet warns about this. A separate rerun with a limit of 300000 iterations gave residual $< 10^{-6}$ after 31015 iterations (gap $\approx 10^{-11}$). For $C = 0.1$ the run stops at the limit only because the residual is 1.05×10^{-6}, just above the threshold, even though the dual gap is already 7×10^{-11}; the same rerun converged in 21775 iterations. So in our runs PG was cheapest at $C = 1$ and more expensive on both sides. For large C the theory gives a partial explanation: the constant in (2) contains $\|\boldsymbol{\alpha}^0 - \boldsymbol{\alpha}^\star\|^2 = \|\boldsymbol{\alpha}^\star\|^2$, which grows by a factor of 6-8 for each step of C in our grid (1.2, 7.1, 56, 387 in Table 2) because most nonzero entries of $\boldsymbol{\alpha}^\star$ sit at the upper bound C, so $\|\boldsymbol{\alpha}^\star\|^2 \approx$ (number at the bound) $\cdot\, C^2$. For small C this does not explain the slowdown, since the bound constant is smallest there while PG is slower than at $C = 1$; we did not find the reason.

- *Barrier cost.* The barrier method is almost insensitive to C: 87-93 Newton steps for $C \geq 1$ and about 180 for $C \leq 0.3$. In the latter two cases the last centering step ($t = 10^{11}$) uses the full 100 Newton steps, because the Newton decrement cannot reach 10^{-9} in floating point. This is the ill-conditioning of the barrier Hessian from Lecture 5: for active constraints its large eigenvalues are of order $\mu_j^{\star 2}/\tau$, so $\kappa = O(1/\tau)$, which is extreme at $\tau = 10^{-11}$ (the last entry of the per-outer-iteration step list is 100). The objective gap (1.9×10^{-9}) and the KKT violation are not affected, so the result is valid, and the extra steps can be avoided by loosening the Newton tolerance.

- *Time.* The barrier method is 17-50 times faster in wall-clock time in all four runs, although each of its steps is much more expensive (a dense 201×201 solve versus a matrix-vector product and a bisection).

6 Personal factor: kernel width γ ($C = 1$)

Table 4: Effect of γ. * PG stopped at the 20000-iteration limit. Eigenvalues below 10^{-8} are treated as numerically zero.

γ	0.5	1	2	5
Support vectors (all three methods)	72	63	59	69
Free SV	5	10	18	35
Training accuracy	0.935	0.940	0.945	0.950
$\lambda_{\max}(Q)$	99.8	68.9	43.5	21.5
Eigenvalues of Q below 10^{-8} (of 200)	136	109	73	13
PG iterations	14001	7302	20000*	7923
PG dual gap	2.0×10^{-12}	3.1×10^{-12}	1.6×10^{-10}	2.0×10^{-11}
PG KKT violation	6.6×10^{-7}	6.0×10^{-7}	1.2×10^{-6}	7.5×10^{-7}
PG time, s	8.86	5.65	12.44	4.80
Barrier Newton steps (outer)	89 (12)	87 (12)	89 (12)	195 (12)
Barrier dual gap	1.9×10^{-9}	1.9×10^{-9}	1.8×10^{-9}	1.6×10^{-9}
Barrier KKT violation	1.2×10^{-10}	3.4×10^{-10}	1.9×10^{-9}	3.7×10^{-10}
Barrier time, s	1.36	0.79	0.22	0.45

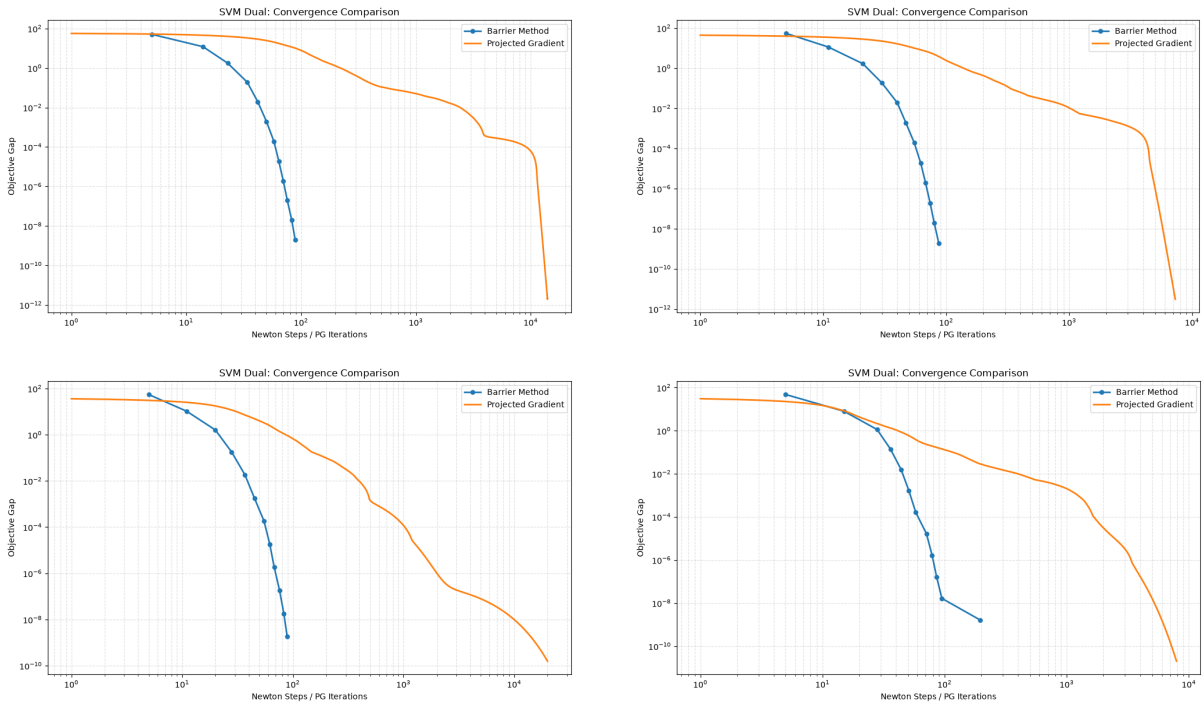


Figure 3: Convergence for $\gamma \in \{0.5, 1, 2, 5\}$ ($C = 1$).

Observations.

- *Boundary complexity.* Small γ gives a smooth, nearly global boundary; large γ makes the kernel local, and at $\gamma = 5$ the boundary wraps around individual points (35 free support vectors versus 5 at $\gamma = 0.5$). Training accuracy increases monotonically ($0.935 \to 0.950$) while the boundary becomes more fragmented. This is consistent with overfitting, but we only measured training accuracy, so we do not claim a loss in generalization.

- *Conditioning of Q.* As γ grows, off-diagonal entries $e^{-\gamma\|x_i - x_j\|^2}$ shrink, Q moves towards the identity, $\lambda_{\max}$ falls from 99.8 to 21.5, and the number of numerically zero eigenvalues falls from 136 to 13. For $\gamma \leq 2$ the matrix is numerically singular (smallest eigenvalue of order 10^{-15}, up to rounding sign), so the condition number $\lambda_{\max}/\lambda_{\min}$ is not meaningful and we report $\lambda_{\max}$ and the number of near-zero eigenvalues instead. At $\gamma = 5$ the smallest eigenvalue is 1.3×10^{-10}, giving $\kappa \approx 1.7 \times 10^{11}$.

- *PG cost.* The iteration count is not monotone in γ: 14001, 7302, $\geq$ 20000 (22531 in an uncapped rerun), 7923. A smaller $\lambda_{\max}$ allows a larger step $1/L$, and the bound constant $L\|\boldsymbol{\alpha}^\star\|^2$ decreases monotonically with γ (6850, 3850, 2020, 940), but the actual counts do not follow it. So the worst-case bound does not predict the trend, and we did not find a simple explanation.

- *Barrier cost.* 87-89 Newton steps for $\gamma \leq 2$. For $\gamma = 5$ the last centering step again used all 100 Newton steps (195 in total), with no loss in accuracy (gap 1.6×10^{-9}).

- *Time.* The barrier method is faster in every case (by factors of about 6-56), with the caveat on timing noise given in Section 7.

7 Limitations and pitfalls

- **Large C with plain PG.** Convergence is very slow already at $C = 3$ (31015 iterations to reach residual 10^{-6} in our rerun, and not reached in 20000). We therefore did not push C

beyond 3.

- **Barrier start.** $\boldsymbol{\alpha}^0 = \frac{C}{2}\mathbf{1}$ satisfies $\boldsymbol{y}^\top \boldsymbol{\alpha}^0 = 0$ only because the classes are balanced (100/100). With unbalanced data a separate feasible start would be needed.
- **Bias.** b is not part of the dual. It is recovered from free support vectors, so at least one free support vector must exist (this was the case in all runs).
- **Timings.** Wall-clock times come from separate runs and vary with machine load. For example, the same configuration $C = 1, \gamma = 1$ took 0.25 s and 0.79 s for the barrier method in two runs. Iteration counts are deterministic and reproducible.
- **Iteration limit.** Runs marked * hit the 20000-iteration limit. For $C = 3$ this means the PG result does not meet the 10^{-6} target (see above).

8 Conclusion

Both methods solve the SVM dual correctly and agree with `SVC` on the objective and on the set of support vectors. In our experiments the barrier method needs about 90 Newton steps independently of C and γ (up to the extra 100 steps in the last centering where it cannot improve further) and reaches a gap of $\approx 2 \times 10^{-9}$. Projected gradient needs 10^4-3×10^4 cheap iterations and is much more sensitive to the parameters, especially to large C. Increasing C narrows the margin and reduces the number of support vectors; increasing γ makes the boundary more local and Q closer to the identity. The PG runs agree with the Lecture 6 theory (monotone decrease, $O(1/k)$ bound satisfied), but the bound is far from tight. A natural next step, not tested here, is FISTA, which has the same cost per iteration and a bound of order $1/k^2$ by Lecture 6.

A Reproducibility

Source code (submitted with this report): `SVM_dual_form.py` (experiment driver, KKT check, plots), `barrier_template.py` (barrier method with objective, gradient, Hessian, constraints, start point and t-update), `projection.py` (projection by bisection and projected gradient). Each configuration is run by setting `C` and `gamma` at the top of `SVM_dual_form.py`. Random seed: `random_state=0`. The extra quantities (training accuracy, margin, spectrum of Q, decision boundaries) are computed with the same data and solutions.